
CE 352: Transportation Engineering

Course Code	CE 352
Course Name	Transportation Engineering
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)
Revised	44th BoA (Nov 2021) - syllabus revised

Course Contents

- **Introduction to transportation engineering**
various methods of transportation and their importance, PMGSY, Golden quadrilateral and other road development plans of GOI. [6 Lectures]
- **Module II**
SGeometrical design, camber, super elevation cross-sectional elements, sight distances, horizontal and vertical alignments, transition curves. [12 Lectures]
- **Highway Engineering**
Pavement material and characterization, aspects of analysis and design of flexible and rigid pavements. [7 Lectures]
- **Highway Planning & Maintenance**
types of failure, evaluation and remedial measures. [7 Lectures]
- **Traffic Engineering** Vehicle and driver characteristics, Traffic design studies. [5 Lectures]
- **Module VI**
Introduction to rail and air transport system, intelligent transport system. [5 Lectures]

Text Books

1. S.K. Khanna and C.E.G. Justo, **Highway Engineering**, Nem Chand Bros., 2002.
2. Kadiyali L.R., **Principles and Practice of highway Engineering**, Khanna Publishers, 1992.
3. Khistry, C.J. and B. K. Lall, **Transportation Engineering – An Introduction**, Prentice Hall of India Ltd., 2003.

References

1. Garber, N.J., Hoel, L.A., **Traffic and Highway Engineering**, West Publishing Company, 2014.
2. P. Chakroborty and A. Das, **Principles of Transportation Engineering**, Prentice Hall India, 2003.
3. S.C. Saxena and S.P. Arora, **A text book of Railway engineering**, Dhanpat Rai, 2001.